Vaccination hot spots

Pneumococcal coverage among adults 65 and over, 2020

Prepared for the Director, CDC · BAN5650 · Group [names]

What we analysed, and what counts as normal

Vaccine	Population	Survey year	Geography
Pneumococcal	**Adults 65+**	**2020**	**State and territory level**

70.3%

National pneumococcal coverage, adults 65+, 2020

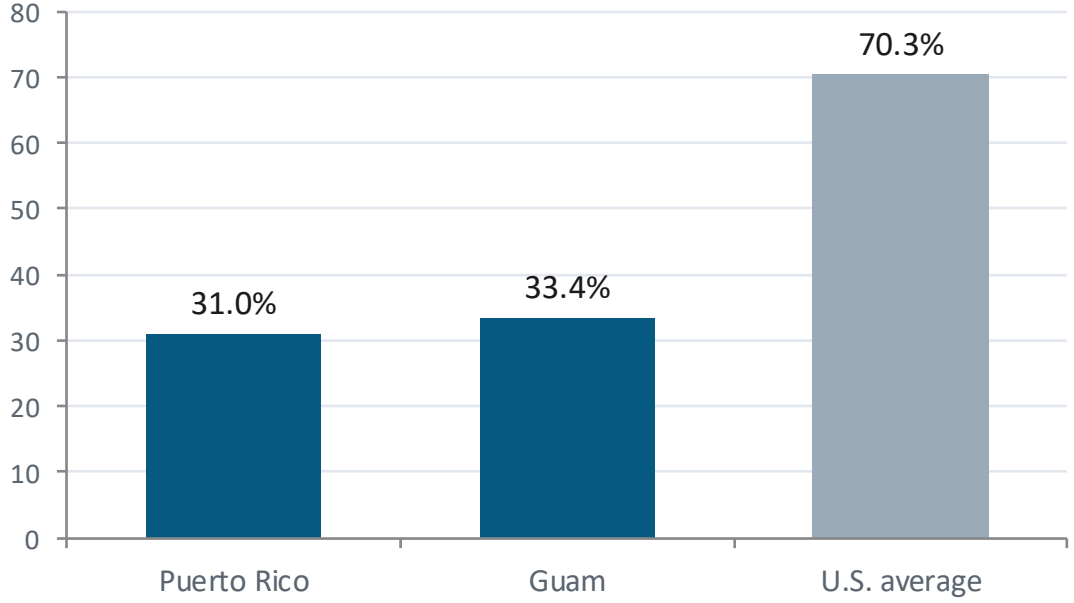
95% confidence interval 69.6 – 71.0 · sample size 124,768

This is the benchmark every finding is measured against.

Hot spot rule: a geography qualifies only if the upper bound of its 95% confidence interval falls below 70.3%, so the shortfall is statistically significant rather than sampling noise. Sub-state areas were excluded so cities are not compared against whole states.

Puerto Rico and Guam are the extreme outliers

Fewer than one in three older adults vaccinated, against seven in ten nationally



31.0%

Puerto Rico

39.3 points below national

n = 1,013 · CI 26.4 – 35.6

33.4%

Guam

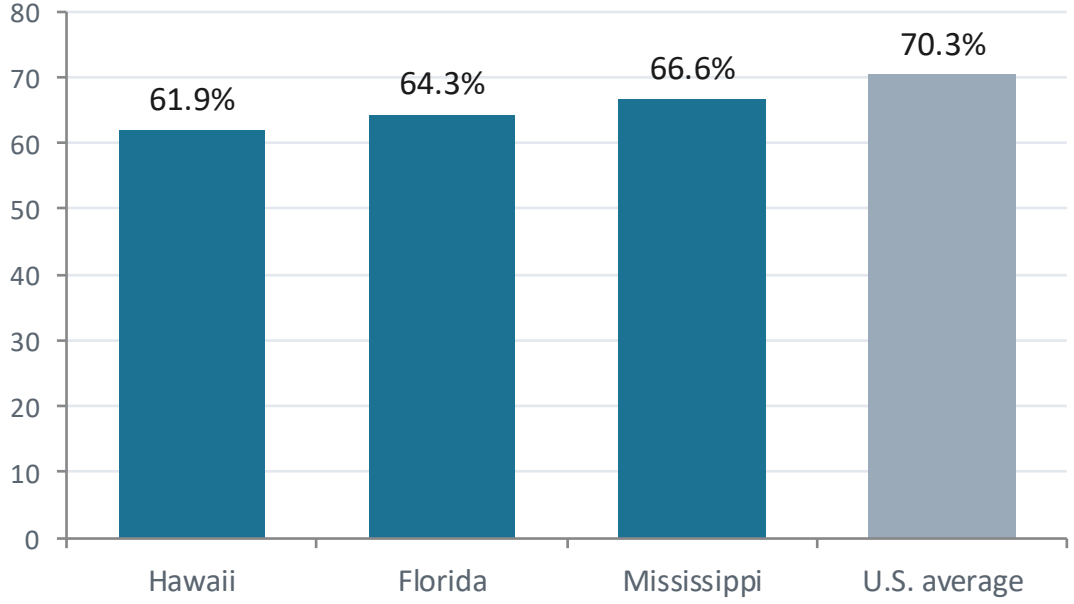
36.9 points below national

n = 433 · CI 23.0 – 43.9

Even the most optimistic reading of Puerto Rico sits 35 points below the national rate.

Three states fall well below national

Each shortfall is statistically significant on a healthy sample



Hawaii **61.9%**

– 8.4 pts

Lowest of the 50 states. n = 2,354

Florida **64.3%**

– 6.0 pts

Largest sample of any low state. n = 4,621

Mississippi **66.6%**

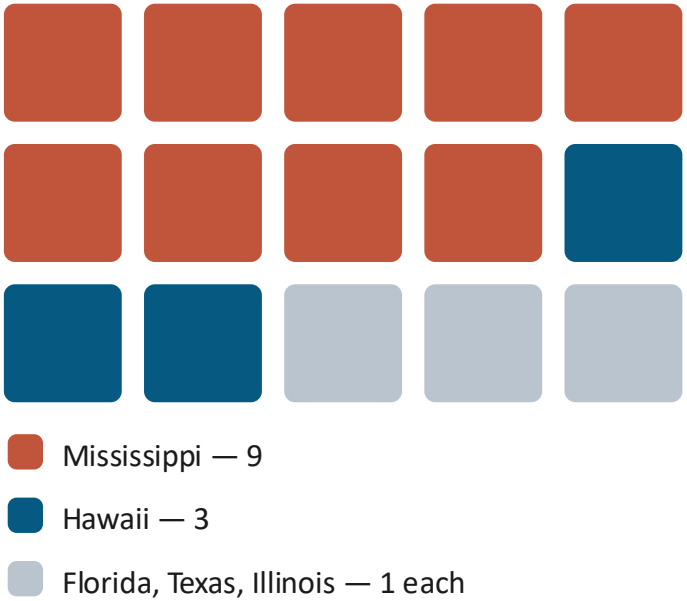
– 3.7 pts

More on this next slide. n = 2,482

All three clear the significance test: the upper bound of each confidence interval sits below 70.3%.

Mississippi holds 9 of the 15 lowest counties

A state with 82 of the nation’s 3,143 counties — 2.6% of the total



Why this matters

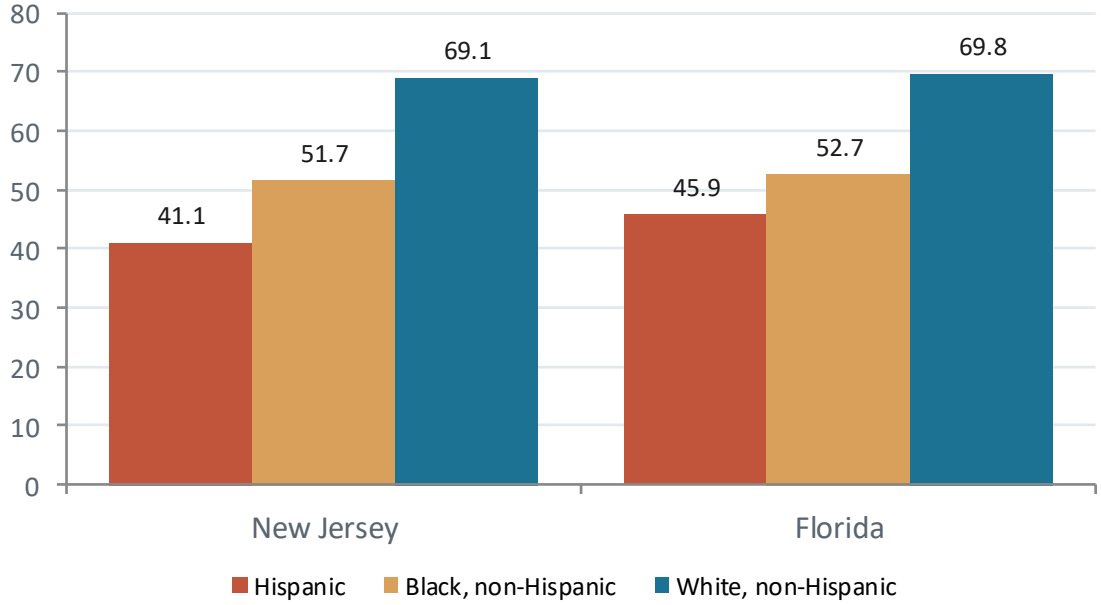
The low counties are adjacent, not scattered — they form a band through the Mississippi Delta. Clusters point to a shared cause; random noise does not.

Three of the remaining six are in Hawaii — Maui, Kauai and Kalawao — which reinforces the previous slide.

This is a count of lowest-ranked counties, not a rate comparison.

Hispanic elders are vaccinated at 41 to 46%

The same pattern appears in both New Jersey and Florida



28.0

point gap in New Jersey

Hispanic 41.1% vs White 69.1%

Baseline shifts here

Race findings are measured against the national White non-Hispanic rate of 73.5%, not 70.3%.

Estimates shown only where the sample exceeds 100 respondents.

This explains part of why Florida appeared two slides ago, rather than repeating it.

Where we would look next

Three of the five hot spots share one characteristic

Puerto Rico	New Jersey	Florida
31.0% overall	41.1% Hispanic	45.9% Hispanic

All three have large Hispanic populations, and that is where we would direct the next round of analysis. *The data shows the gap; it does not yet show the cause.*

Source: CDC adult vaccination coverage, BRFSS. Pneumococcal, adults 65+, survey year 2020.